

Samsung Z2 (Tizen 2.4) Development Environment Setup Guide

Project: Telescreen

Purpose

Document the complete development environment required to build and deploy native applications to the Samsung Z2 (SM-Z200F) running Tizen 2.4.0.7. The guide captures the working configuration established during the project so it can be reproduced in the future.

Target Device

Model: Samsung Z2 (SM-Z200F)
OS: Tizen 2.4.0.7
Kernel: Linux 3.10.65 (ARMv7)
Build: Z200FDDU0BQL1
User: developer (via sdb)

Required Hardware

- Windows 11 PC
- Samsung Z2
- Micro-USB DATA cable (charge-only cables will not work)
- Internet connection
- USB port (USB 3.x ports are acceptable)

Developer Options

1. Enable Developer Options.
2. Enable USB Debugging.
3. Connect the phone using a data-capable USB cable.
4. Accept the debugging authorization dialog and select 'Always allow from this computer'.

SDK Installation

1. Install Tizen Studio 3.7.
2. Launch Package Manager.
3. Disable 'Auto Update'.
4. Legacy baselines (Tizen Studio 2.3/2.4) become visible.
5. Select Tizen Studio 2.4.
6. Install Mobile 2.4 Native SDK and dependencies.

USB Driver Troubleshooting

Symptoms encountered:

- Phone charging but not detected.
- Cause: charge-only USB cable.
- After replacing cable, Windows detected 'Samsung SDB' with a yellow icon.
- Installing the Samsung USB driver resolved the issue.

Verification Commands

```
sdb version  
sdb devices
```

```
sdb shell uname -a  
sdb shell cat /etc/info.ini  
sdb shell id  
sdb shell whoami
```

Expected Results

Smart Development Bridge 2.3.8
Device listed as 'SM-Z200F'
Kernel: Linux 3.10.65
User: developer

Installed Tools

SDK Root: C:\tools\tizen_sdk_3.7
sdb.exe: tools\sdb.exe
package-manager-cli.exe: package-manager\package-manager-cli.exe

Known Issues

- 'sdb capability' returns 'closed' on this SDK and can be ignored.
- 'sdb root on' returns 'Permission denied' on production firmware (expected).
- Modern VS Code Tizen extension targets Tizen 8/9/10 and is unsuitable for the Z2.
- Use the legacy Mobile 2.4 SDK instead.

Recommended Workflow

VS Code as the editor.
Use Git and GitHub for version control.
Use the legacy SDK only for compilation, packaging and deployment via sdb.
Keep the project portable by implementing core functionality in a reusable C++ library (libminiadb) with a thin Tizen-specific wrapper.

Status

The Samsung Z2 is now a fully functional native development platform ready for the Telescreen project.